CVT ESTIMATOR — SOFTWARE REQUIREMENTS SPECIFICATION

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REVISION HISTORY

Rev	Date	Change
A	2025-12-15	Initial release: §1–§6, §7.1, §7.2, §7.4.
B	2026-04-20	Adds §7.3 measurement validation gate for v2.3. Threshold fixed in this document at 5.991 (chi-squared, 95 % confidence, m = 2).
C	2026-09-08	§7.3 revised for v2.4: the threshold becomes a configuration value that depends on the confidence level and on the measurement dimension of the active mode. Adds Doppler mode (§2, §6). Adds Appendix A, the fail-closed rules of §7.3.5 and requirements G1–G5.

1. SCOPE

This specification defines software requirements for the Constant-Velocity Tracker (CVT) state estimator and its measurement-processing front end. The core recursion (predict/update) is specified separately in CVT-EST-001; this document specifies the surrounding measurement-processing behaviour.

The implementation language is MATLAB (executed on GNU Octave for validation).

2. DEFINITIONS

state	x = [px; py; vx; vy] (position and velocity, 2-D)
measurement	z = H x + noise; its dimension m depends on the mode
planar mode	position-only sensor: z = [px; py], H = [I2 0], m = 2
Doppler mode	position and velocity sensor (v2.4 onward): z = [px; py; vx; vy], H = I4, m = 4
innovation	y = z - H x (measurement residual, m-by-1)
innovation covariance	S = H P H' + R (m-by-m)
NIS	normalized innovation squared (see 7.3.1)
gate	a test that admits or rejects a measurement before it is applied in the update

3. APPLICABLE DOCUMENTS

CVT-EST-001	State Estimator Specification, Rev C (2026-03-02)
CVT-VAL-002	Validation Plan, Rev A (2026-05-12)

4. MEASUREMENT INTERFACE

Measurements arrive as the rows of an N-by-m matrix Z, where m is the measurement dimension of the active mode. Row k is z_k. The estimator processes them in order. Timestamps, framing and transport are out of scope for this document.

The active mode is selected by configuration (estimator_config.json, key active_mode). A campaign run may override it per run.

5. STATE INITIALISATION

Initial state x0 and covariance P0 are supplied in model.json and are not redefined here. See CVT-EST-001 §2.3.

6. PROCESS AND MEASUREMENT MODELS

The process model F and process noise Q are as defined in CVT-EST-001. The measurement model H and measurement noise R are per mode:

planar	H = [I2 0] (2-by-4), R from model.json (2-by-2)
Doppler	H = I4 (4-by-4), R from the mode entry in estimator_config.json (4-by-4)

This document adds no new constants to the process model.

7. MEASUREMENT PROCESSING

7.1 Innovation computation

For each measurement the front end forms the innovation y and its covariance S from the current predicted state, exactly as in CVT-EST-001 §4.2. These are the inputs to the gate of §7.3.

7.2 Ordering

Measurements are gated in arrival order. Gating has no memory between measurements; each decision depends only on that measurement's (y, S) and the threshold in force.

7.3 MEASUREMENT VALIDATION GATE

Rationale. On 2026-02-18 (validation batch val-2026-02-18, test 1847) a truncated input file put a non-finite measurement into the filter, and the non-finite value propagated through the state. CVT-EST-001 Rev C added requirement R6: non-finite or malformed measurements shall be rejected and reported, never propagated into the state. This gate implements R6 and also screens statistically implausible measurements.

Rev B of this document introduced the gate for v2.3.0 with a fixed threshold of 5.991 (95 % confidence, m = 2). Rev C supersedes that: with Doppler mode the measurement dimension is no longer always 2, so the threshold is a configuration value (7.3.3).

7.3.1 Normalized innovation squared (NIS)

For a measurement with innovation y (m-by-1) and innovation covariance S (m-by-m, symmetric positive definite), the normalized innovation squared is

```
d2 = y' * inv(S) * y
```

Compute d2 by solving S x = y (x = S \ y in MATLAB) and forming d2 = y' * x. Do not form inv(S) explicitly.

7.3.2 Gate decision

The gate admits the measurement if and only if

```
d2 <= gamma
```

where gamma is the gate threshold (7.3.3). A measurement with d2 exactly equal to gamma is admitted. If d2 > gamma the measurement is rejected as a statistical outlier.

7.3.3 Gate threshold

Under the hypothesis of a valid measurement, d2 is chi-squared distributed with m degrees of freedom. gamma is the chi-squared critical value for m degrees of freedom at the configured confidence level, where m is the measurement dimension of the active mode (Appendix A). The confidence level is a configuration item; the default is 99 %.

gamma is resolved from the configuration by the caller and supplied to the gate as a parameter. The gate itself holds no threshold value.

7.3.4 Required interface

The gate is this MATLAB function:

```
[accept, d2] = measurement_gate(y, S, gamma)
```

`y`	m-by-1 innovation vector.
`S`	m-by-m innovation covariance (symmetric positive definite).
`gamma`	scalar gate threshold (a chi-squared critical value).
`accept`	logical scalar: true iff the measurement passes the gate.
`d2`	scalar normalized innovation squared (7.3.1).

The function SHALL be pure: no file access, no printing, no globals. It SHALL work for any measurement dimension m.

7.3.5 Boundary conditions (fail closed)

G-BC-1 If any element of y or S is non-finite (NaN or +/-Inf), or gamma is not a finite positive scalar, the gate SHALL reject the measurement: return accept = false and d2 = Inf. A malformed measurement or threshold must never lead to a measurement being admitted.

G-BC-2 The gate SHALL NOT raise an error for the inputs of G-BC-1; it returns the rejecting result of G-BC-1.

7.3.6 Requirements summary

G1 Compute d2 per 7.3.1 (solving S x = y; not forming inv(S)).
G2 Admit iff d2 <= gamma (7.3.2).
G3 Use the supplied gamma (7.3.3); the gate holds no threshold of its own.
G4 Fail closed on non-finite y or S and on a non-finite or non-positive gamma (G-BC-1), without error (G-BC-2).
G5 Be accompanied by a runnable test that exercises G1–G4.

7.4 LOGGING

Rejected measurements may be logged with their d2 for later analysis. Logging format and transport are out of scope for this document.

8. VERIFICATION

The gate is verified by exercising measurement_gate over recorded innovation streams from both modes, confirming d2 and the accept/reject decisions against this specification, and confirming the boundary behaviour of 7.3.5.

APPENDIX A — CHI-SQUARED CRITICAL VALUES

Critical value gamma of the chi-squared distribution, by degrees of freedom (DOF = measurement dimension m) and confidence level:

DOF	95 %	99 %	99.9 %
1	3.841	6.635	10.828
2	5.991	9.210	13.816
3	7.815	11.345	16.266
4	9.488	13.277	18.467

Measurement dimensions of the defined modes are given in §2.

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